

Differentiable forward models for hologram correction

Yiling Q. Li¹, Congli Wang², Felix Heide²

¹Princeton University Department of Electrical and Computer Engineering

²Princeton University Department of Computer Science

Objective

- The quality of holograms often suffer from experimental and simulation mismatch due to imperfections along the optical path
 - Wavefront distortion, misalignment, nonideal SLM response, etc.

How can we use computational methods to correct for the observed experimental and simulation mismatch?

Introduction to Holography

- **Holography**
 - Technique where wavefronts can be recorded then reconstructed
- **Computer-generated holography (CGH)**
 - Apply computer algorithms to generate or prepare holograms
- **Metasurface holography**
 - Use of subwavelength nanostructures
 - Design geometry of nanoposts to shape light wavefront for hologram generation^{[1],[2],[3]}
- Applications: AR/VR display, medical imaging, optical tweezers, 3D printing, etc.^{[4],[5]}

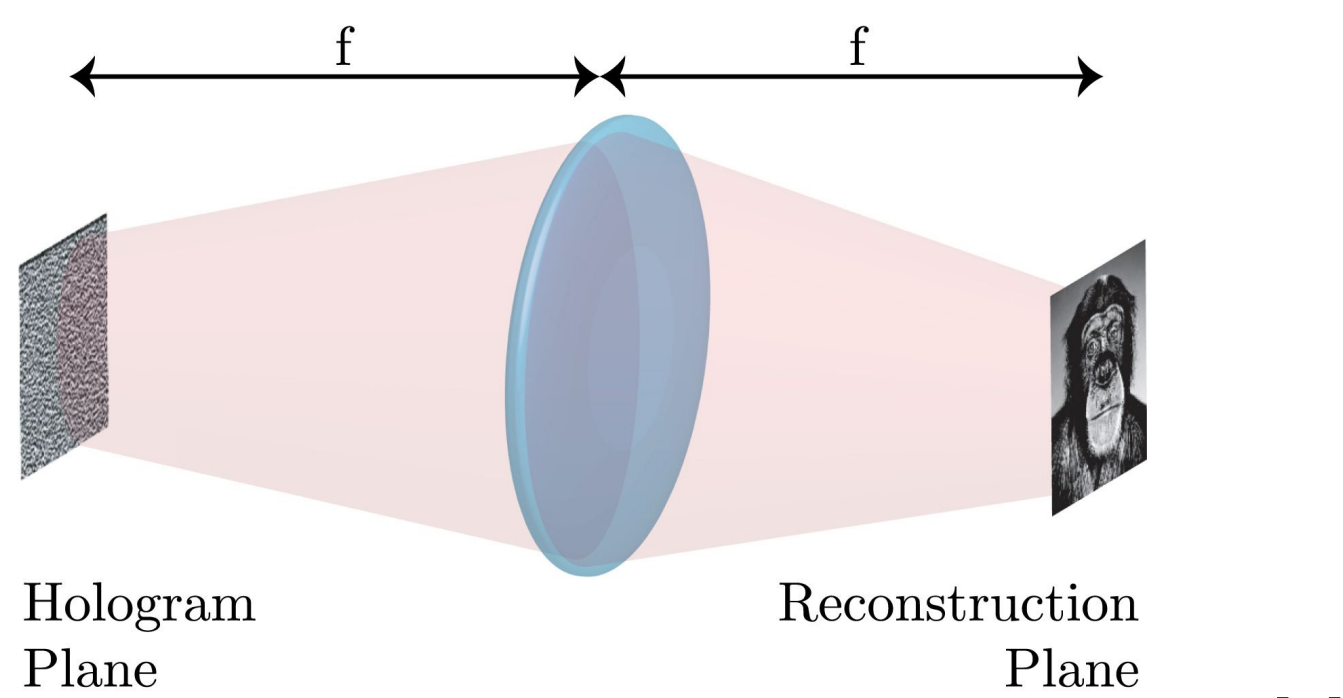


Figure 1: Holography optical setup [6]

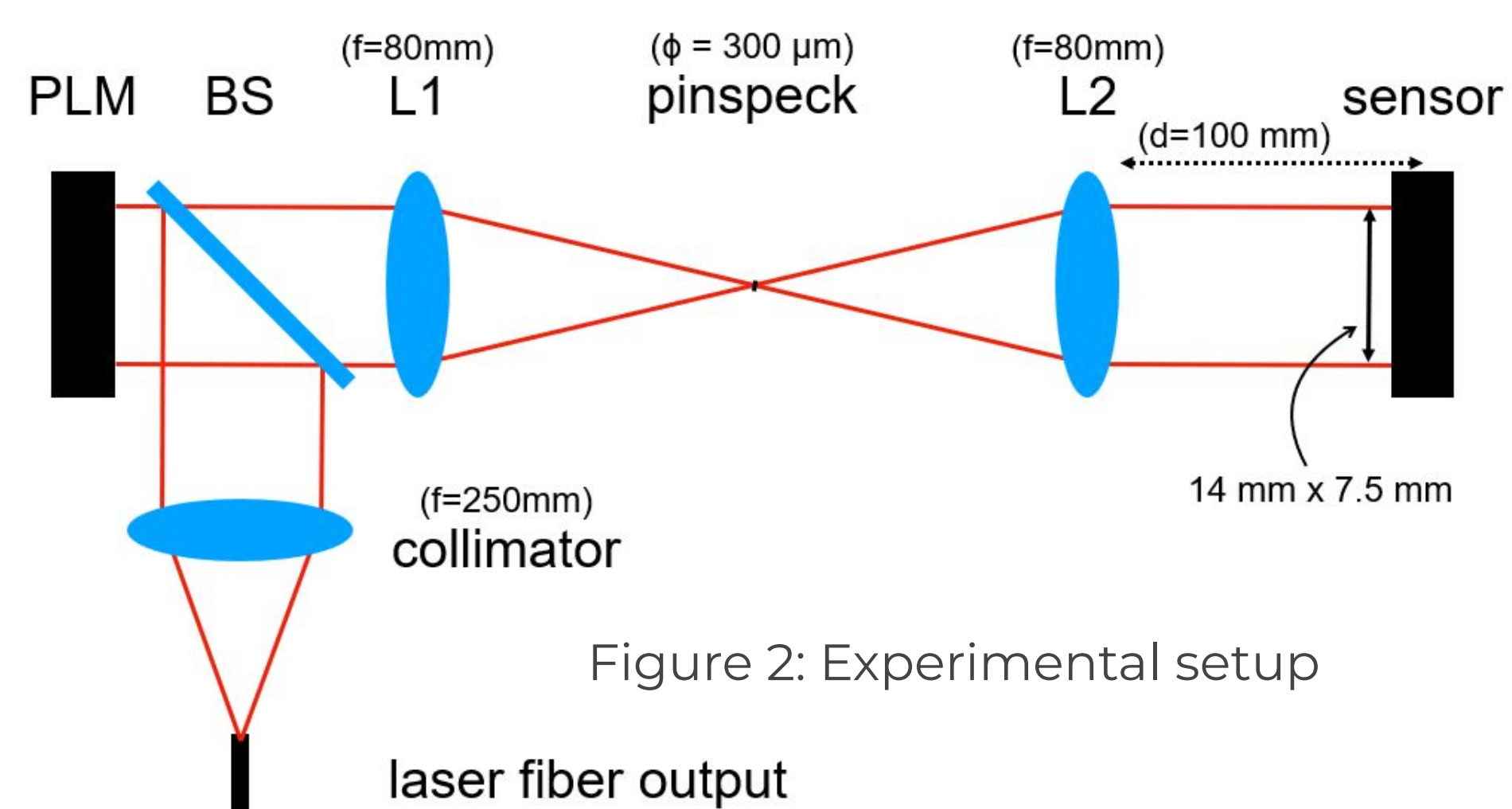


Figure 2: Experimental setup

Training Methodology

- Train model
 - 2k dataset
 - Shapes
 - Borders
 - Images

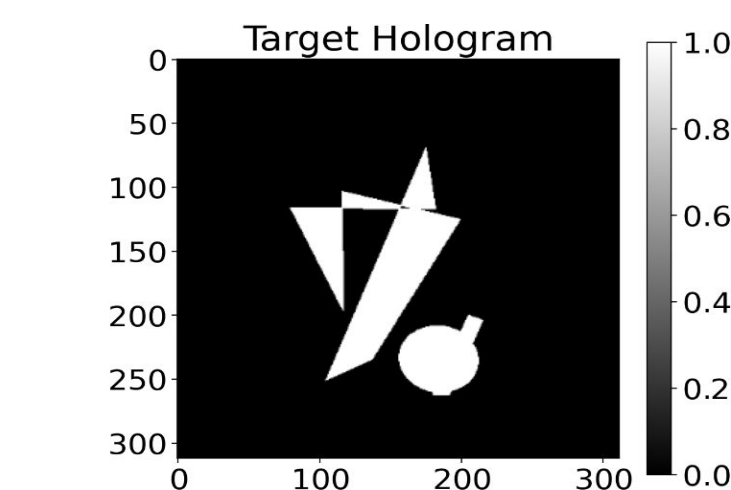


Figure 3: Dataset sample

- Camera-in-the-loop (CITL)
 - Captured hologram is compared with simulated hologram^{[7],[8]}
 - Update phase in direction to minimize loss

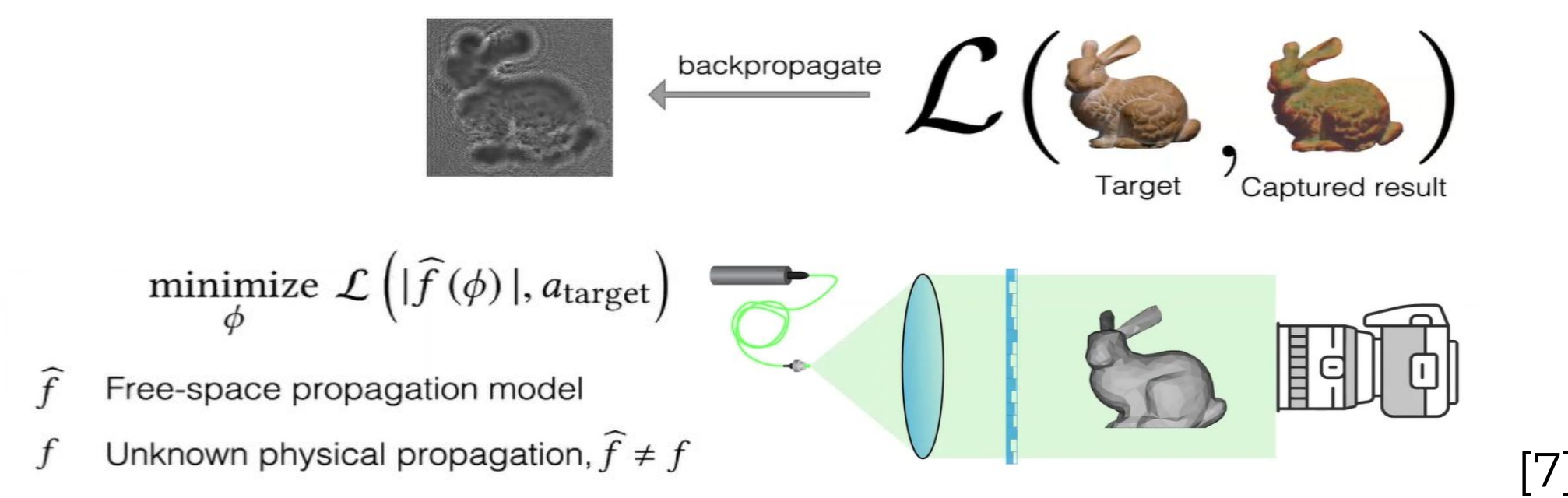


Figure 4: CITL approach [7]

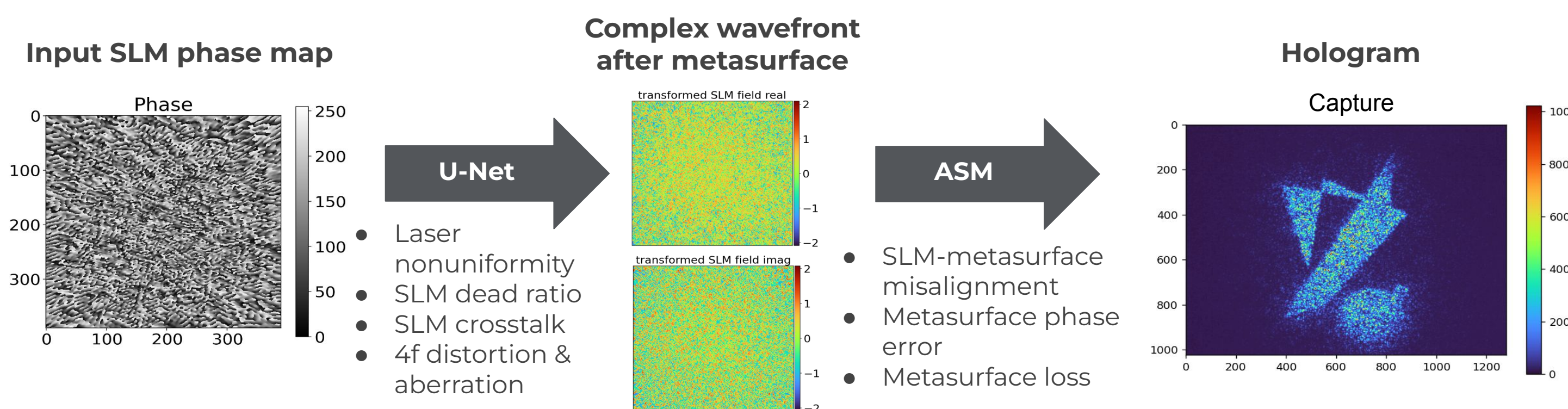


Figure 5: Algorithm progression and potential errors

PLM Proxy Model Results

- Proxy-in-the-loop optimization yields +2.58 dB PSNR / +0.109 SSIM over naive ASM
- Learned Zernike coefficients compensate for aberrations and quantization loss
- Per-wavelength proxy tuning balances convergence across RGB channels
- Only ~100 simulated (phase, capture) pairs needed to train the 9-parameter proxy
- Physics backbone (ASM) ensures generalization without large datasets



Figure 6: Qualitative comparison for image 000001

Metasurface System Results

- Trained model with CITL and XYZ autofocus performs the best
- Trained model & CITL account for physical characteristics
 - Correct for residual error not captured in pure simulation
- CITL model can learn and correct for speckle (grainy, noise-like patterns)
- Partially corrects for zeroth-ordered light (non-diffracted light)

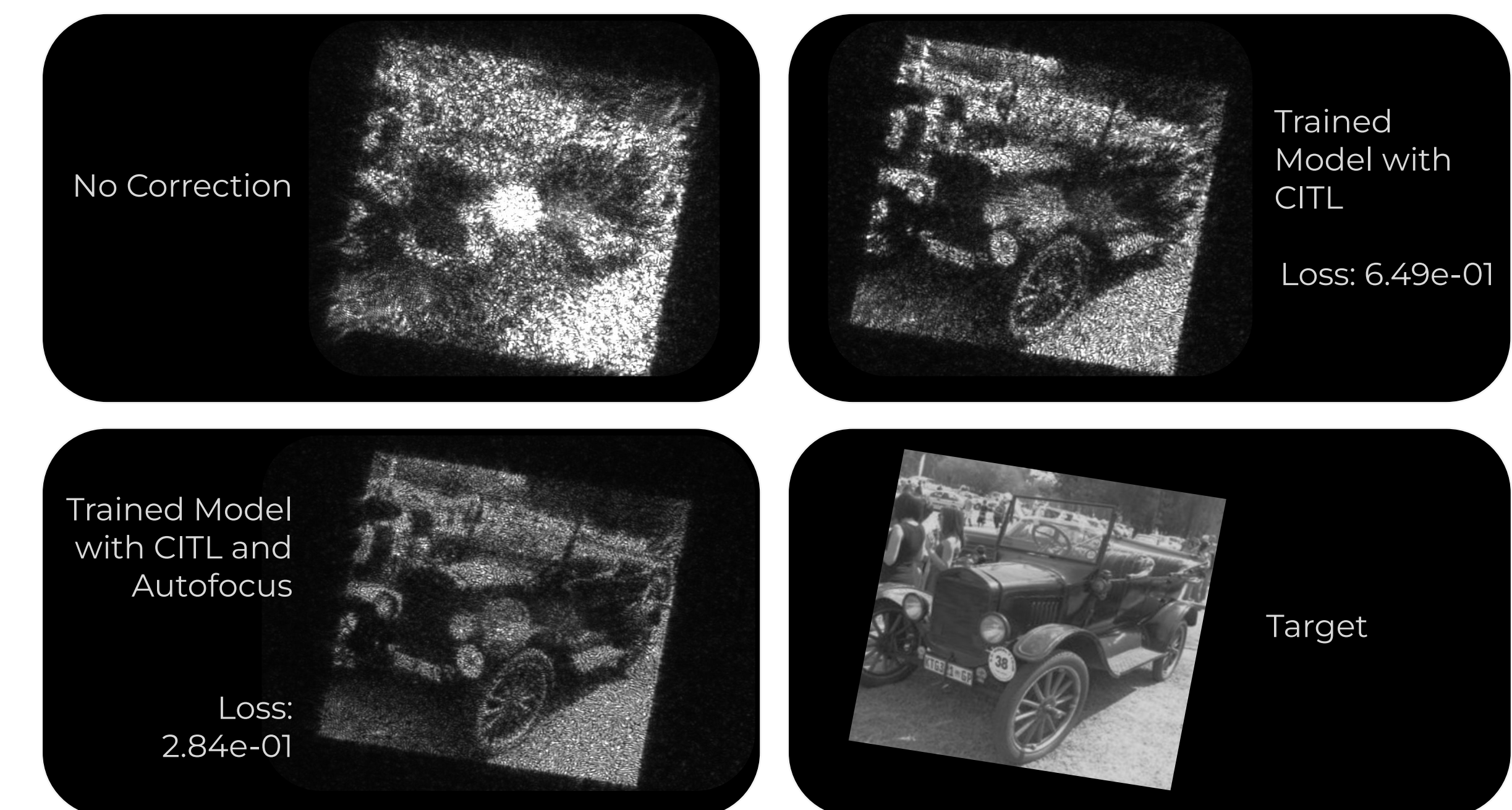


Figure 7: Comparison of hologram captures

Future Directions

- Extend model to generalize across different metasurface designs
- Add adaptive learning rate scheduling for both proxy training and CITL
- Scale pipeline to multi-plane (3D) holographic reconstruction
- Integrate neural étendue expansion to expand FoV beyond SLM aperture

Acknowledgements & References

I would like to thank Congli Wang, Victor Chu, and Yu Wu for providing research supervision and building the optical experiment test beds along with Professors Felix Heide and Jonathan Fan for advising this project. I would also like to thank Chenkai Mao for helping with the neural network model and Sara Azzouz for fabricating the metasurfaces. This project was made possible with funding from the Princeton SEAS, Princeton ECE, and Stanford EE departments.

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